Windows 10 Setup

Objectives

1. Create an ISO image of the Windows 10 machine.
2. Install and configure Windows 10 machine for use.
3. Setup static IP address in Windows 10.
4. Perform ping test between Kali and Windows VMs.

VM Overview

Purpose	Workstation
Base OS	Windows
Installed On	VirtualBox 7.1.4
Resources Allocated	2 CPUs, 2 GB RAM, 30 GB Disk
Network Mode	Internal Network (LabNet)

Installation Process

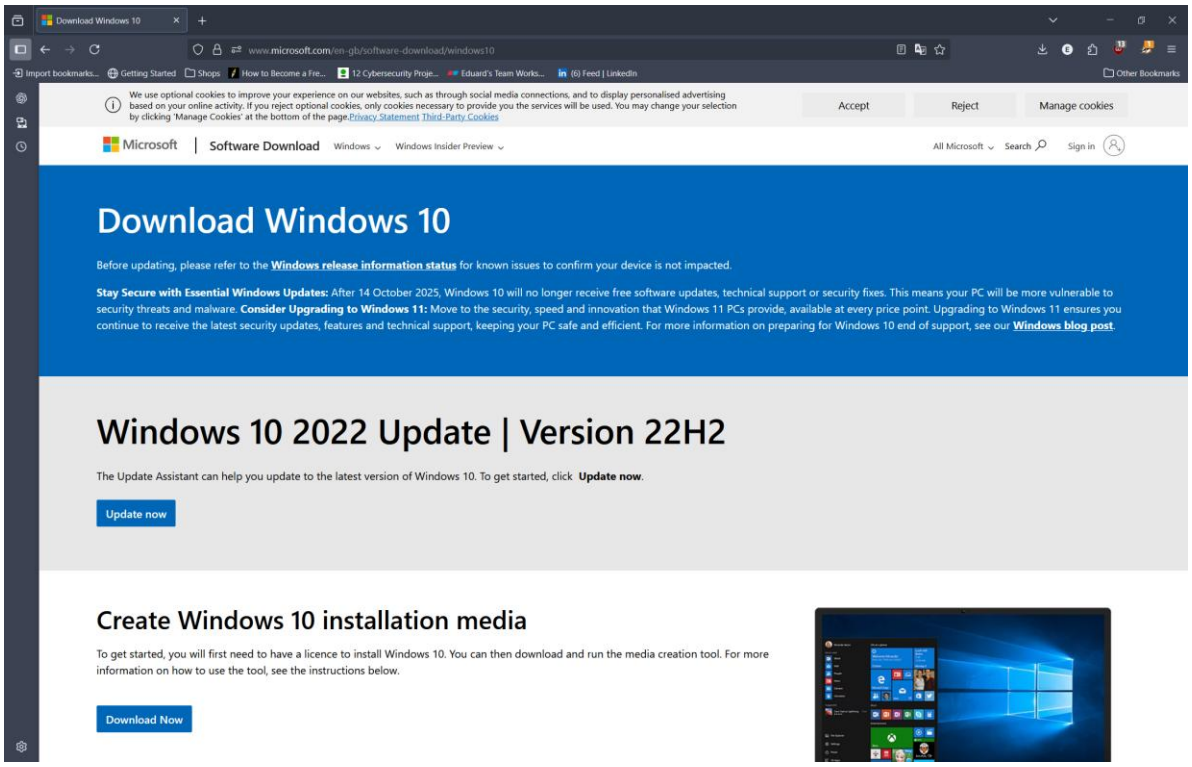
ISO Download

- Source: https://www.microsoft.com/en-gb/software-download/windows10
- ISO File: Windows10.iso

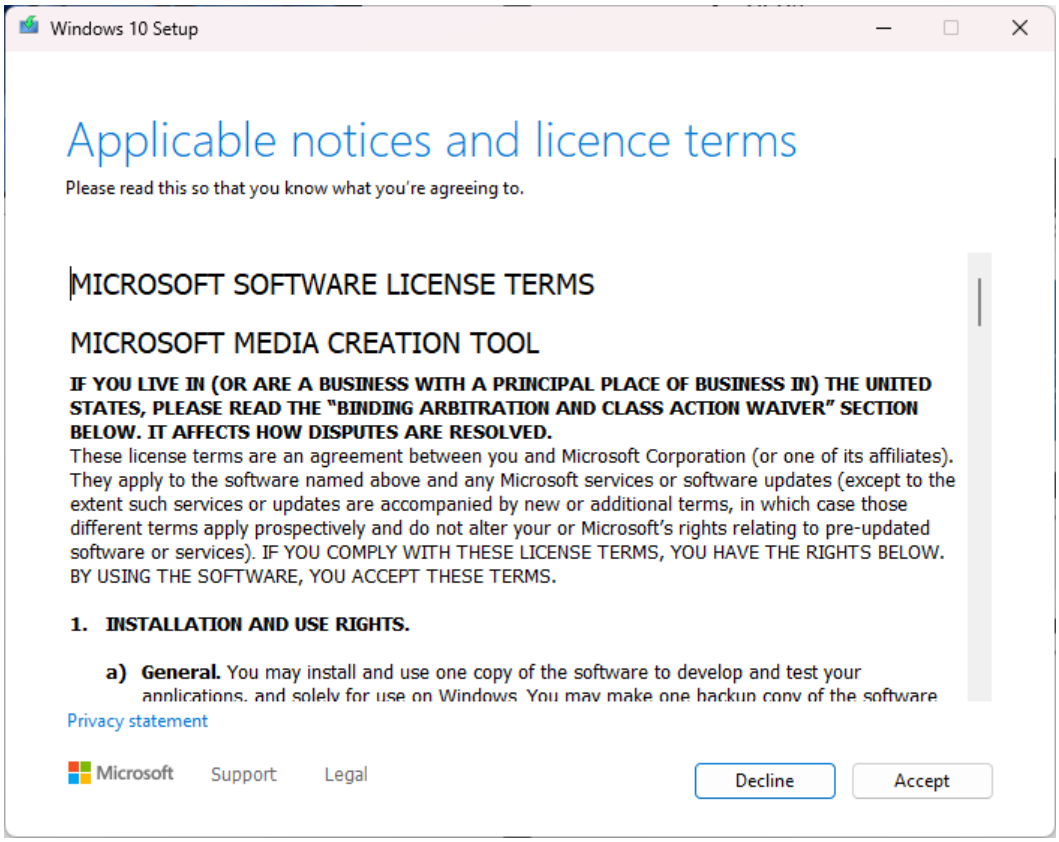
Virtual Machine Configuration

- Hypervisor: VirtualBox
- Name: Windows10
- Type: Windows
- Base Memory: 4096 MB
- CPUs: 2 CPUs
- Network Adaptor: NAT (It will be changed later)
- Hard Disk: 50 GB

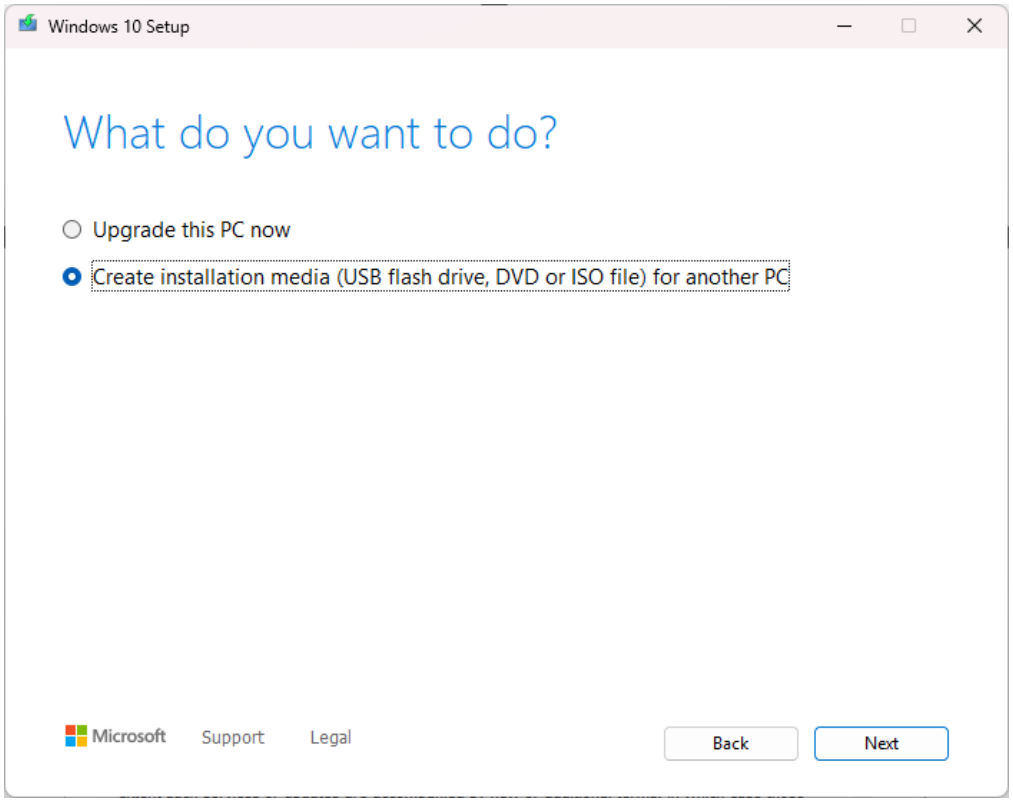
Screenshots



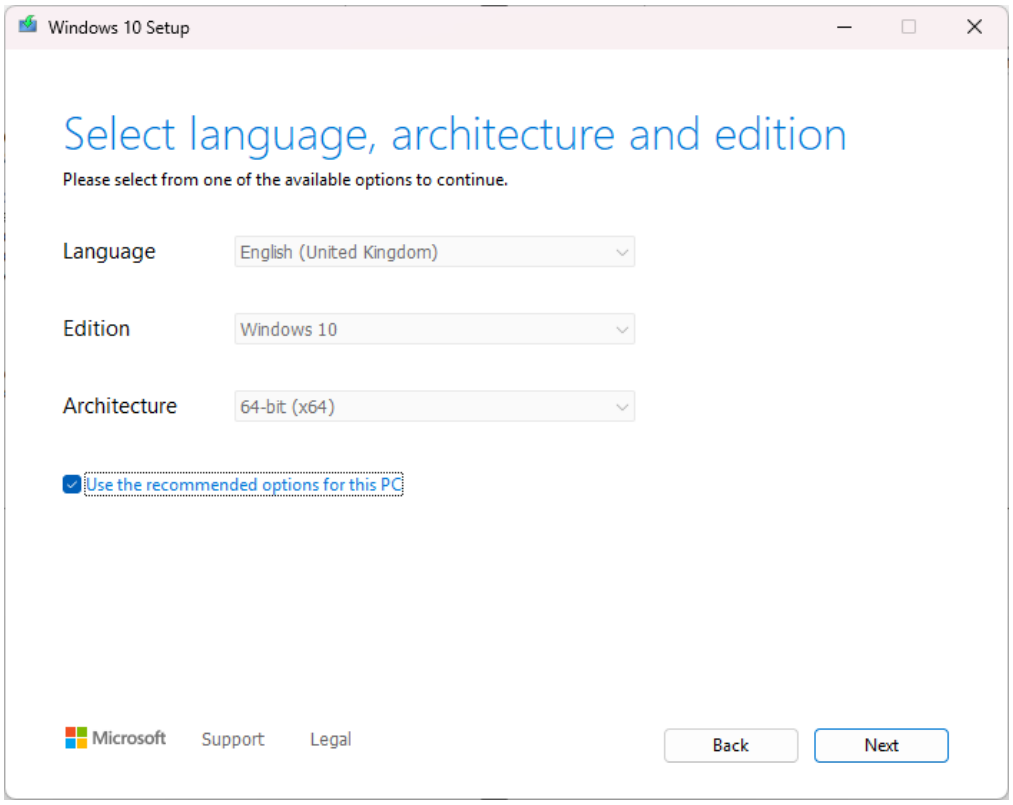
On official Microsoft page you will be able to download a windows 10 installation media. This will allow us to create an windows 10 iso file.



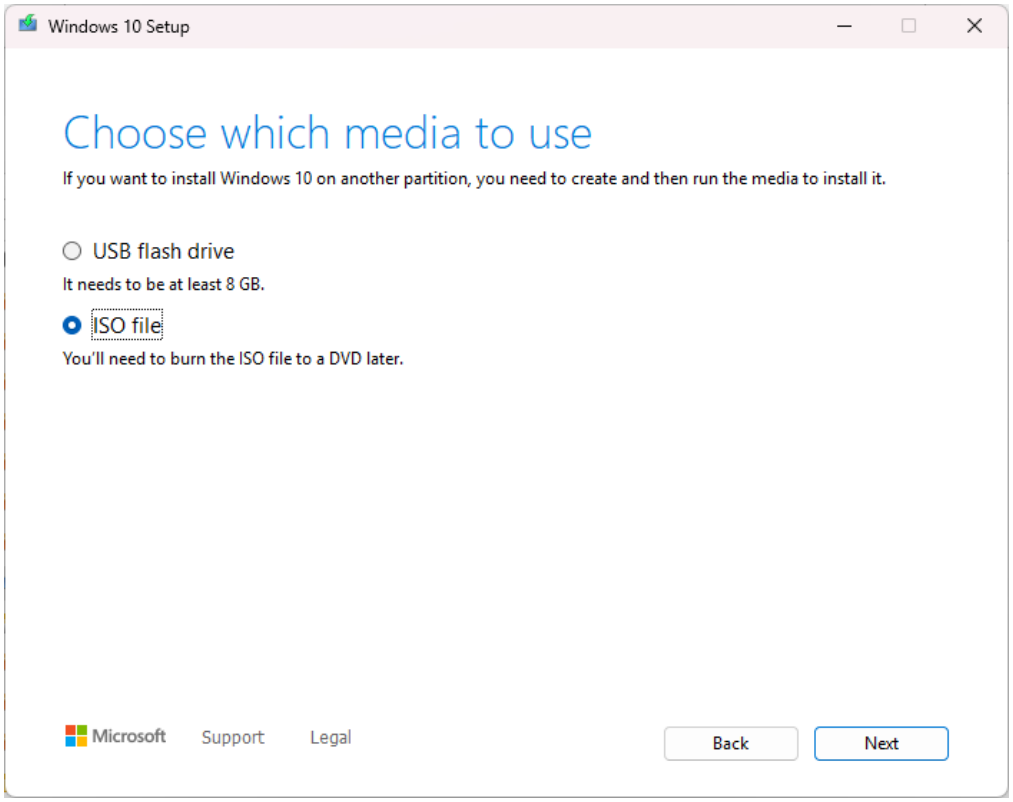
Run the downloaded exe file, then accept the license terms.



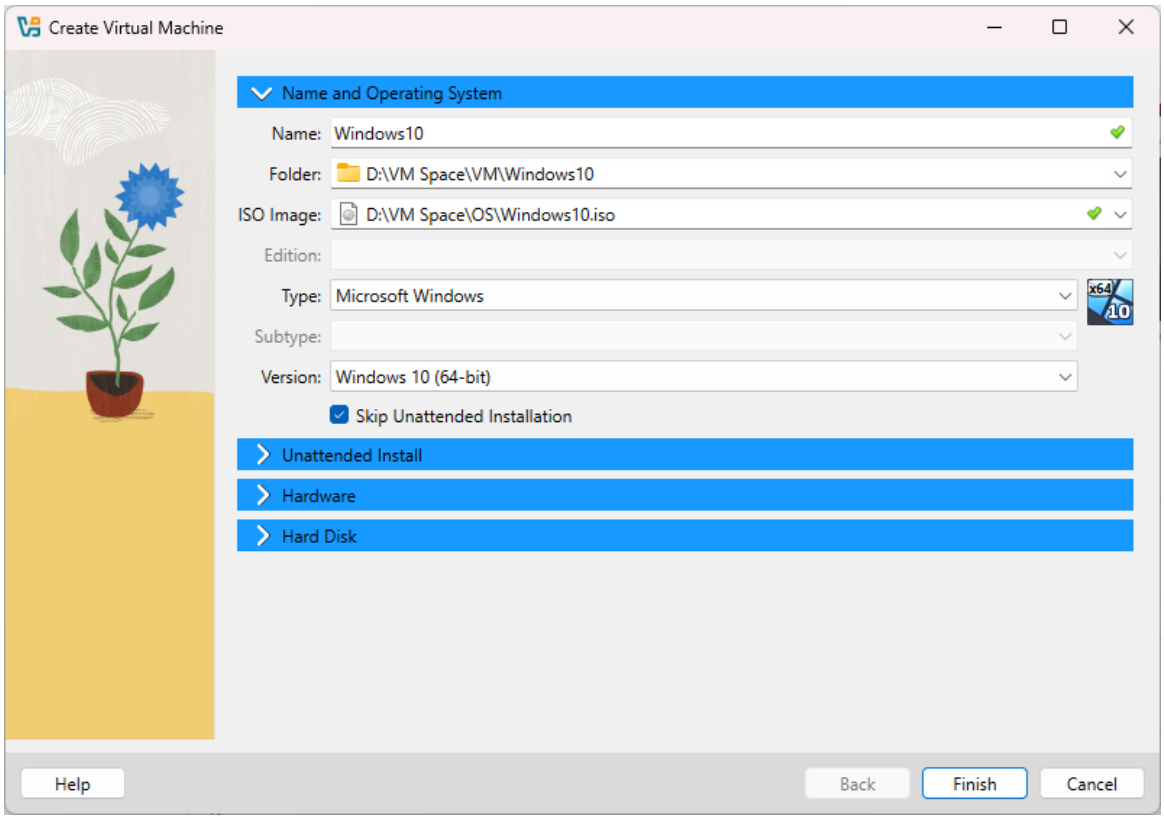
Then choose create installation media, which can be used to create windows 10 on USB flash drive, DVD or ISO file.



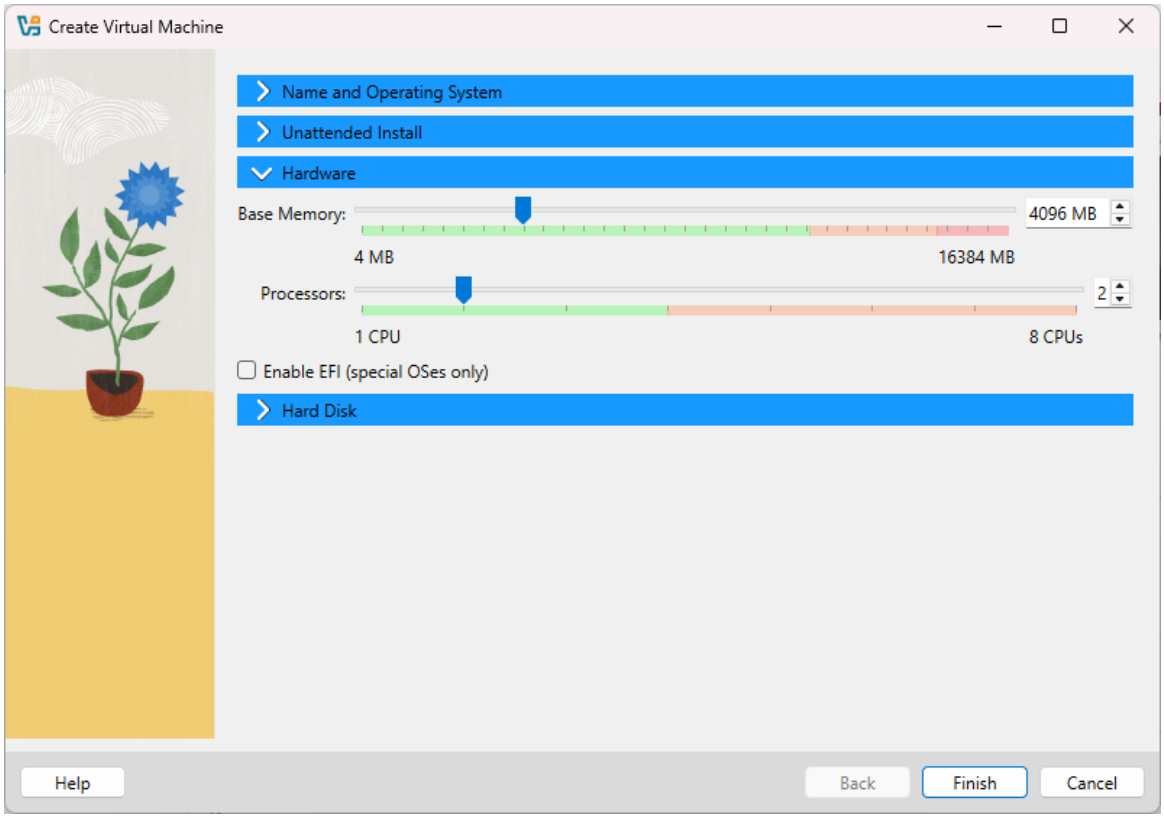
For this step you can adjust settings according to your PC setup. In my case, I enabled recommended options for my PC.



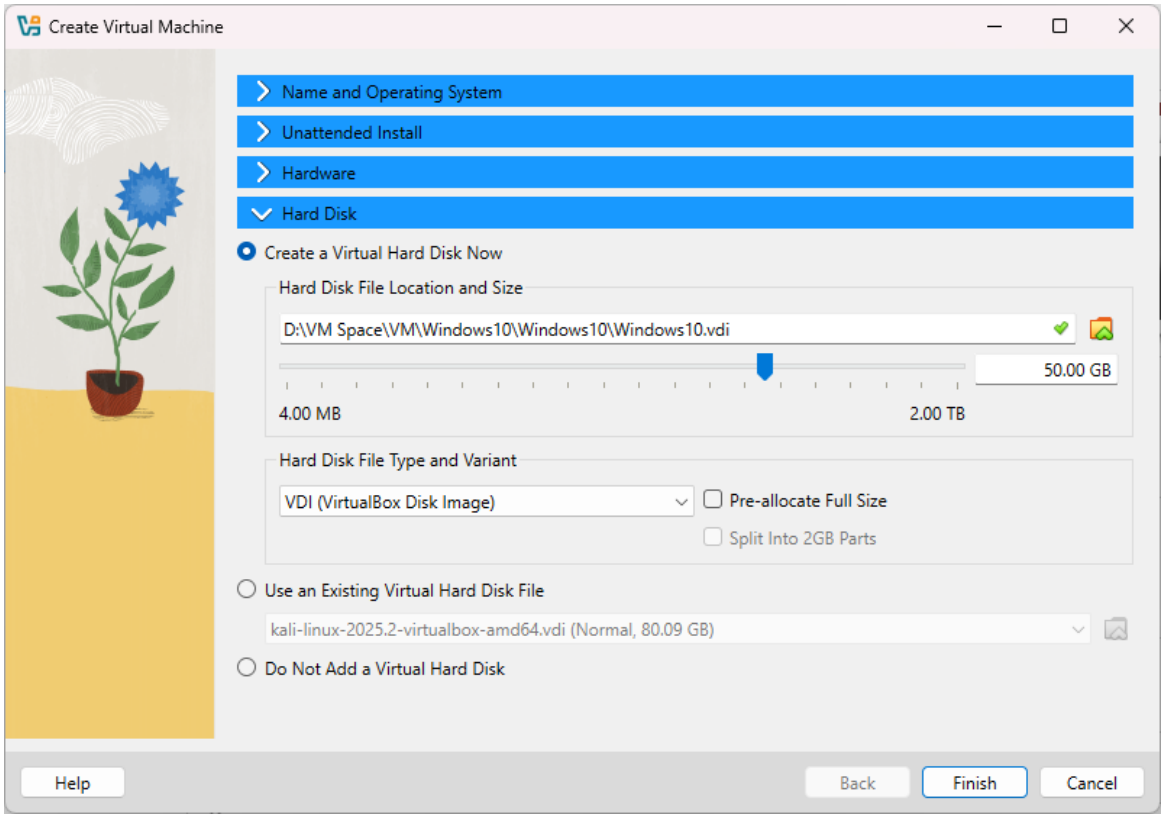
Then choose ISO file and click next. This should create a ISO installation file for Windows 10.

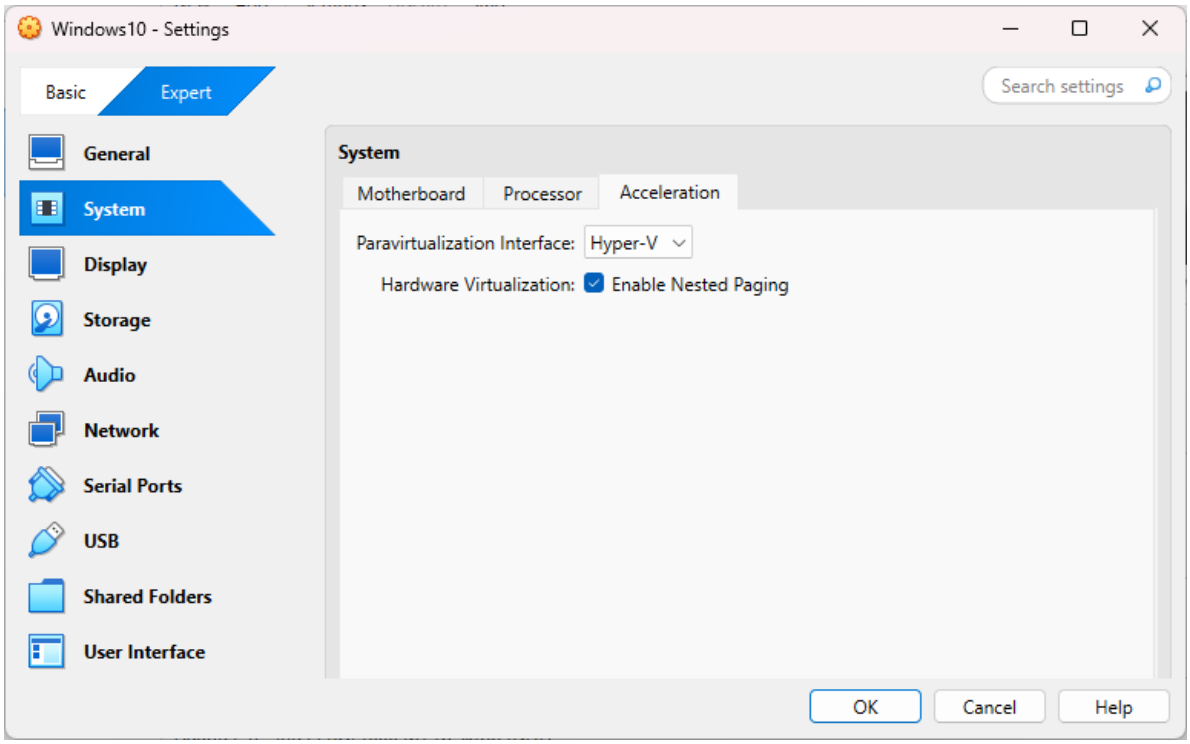


Then in virtual machine select the windows 10 iso file and the folder you want to store the virtual machine.

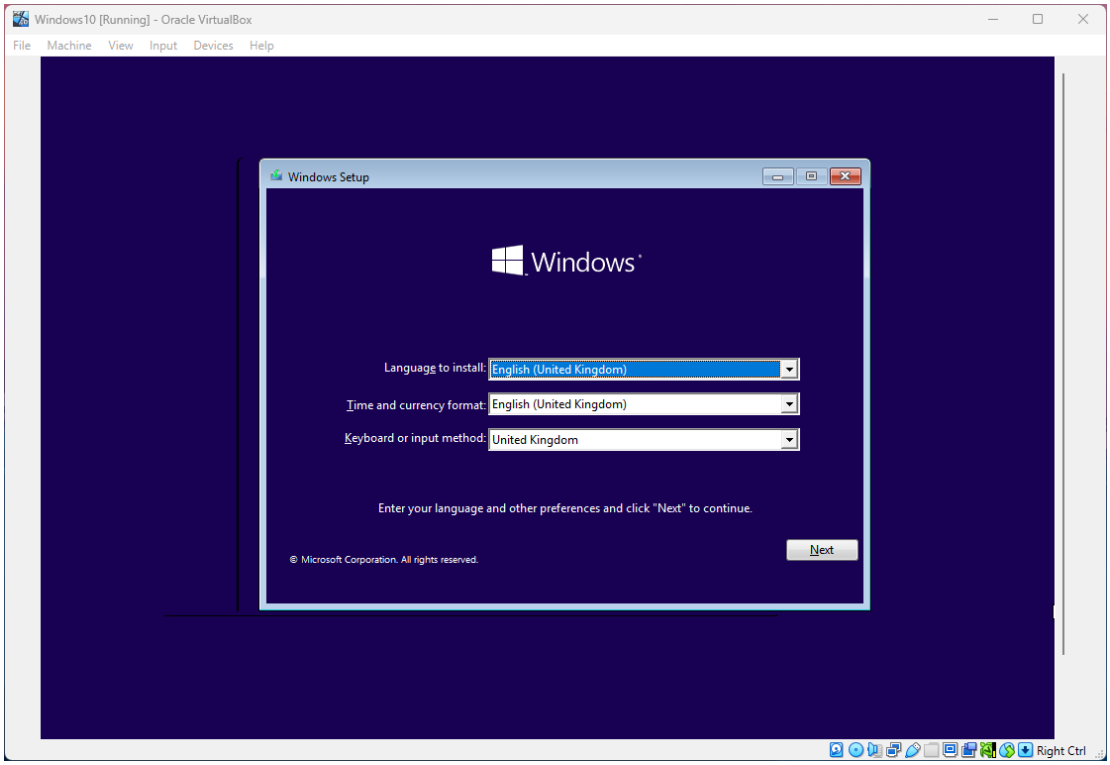


Since I know the Windows 10 is resource intensive I allocated more base memory to it, around 4 GB and allocated 2 CPUs to it. While it may not be enough but for testing, I think it fine. Bellow I selected 50 GB of Hard Disk due to windows 10 being intensive on the storage too.

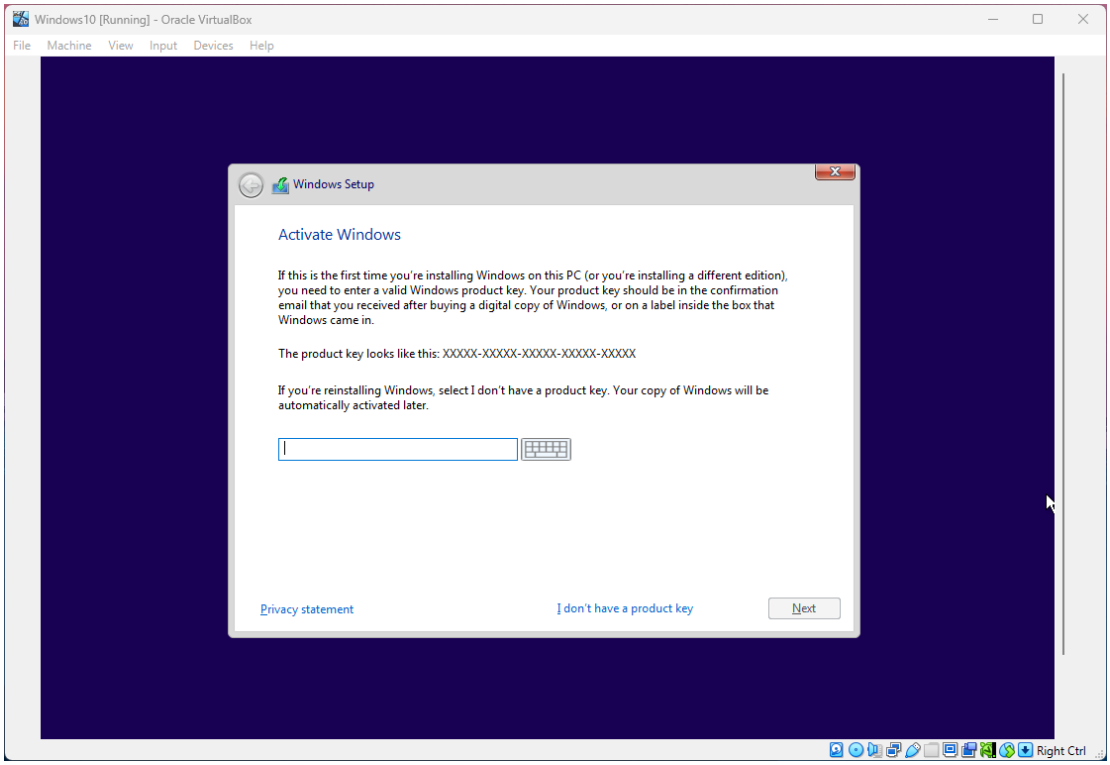




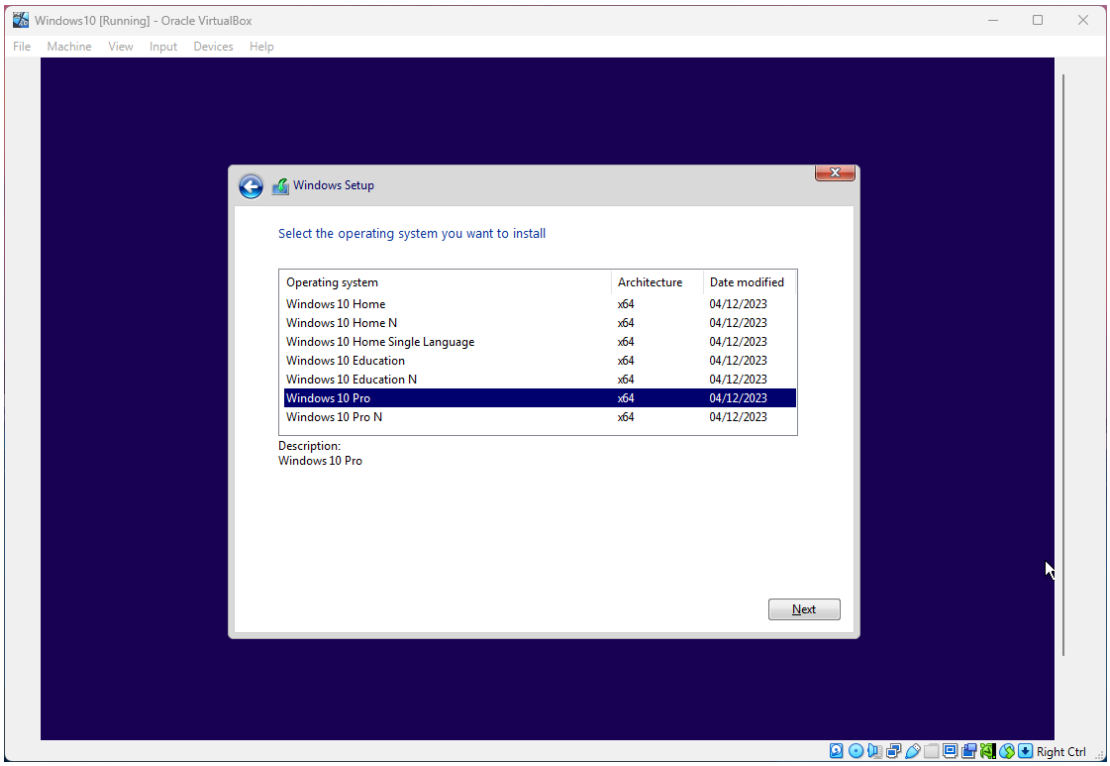
For this virtual machine, go to settings, then go to system and go acceleration section. For the paravirtualization interface select Hyper-V option. This option works quite well with windows operating systems. Once its done, run the virtual machine.



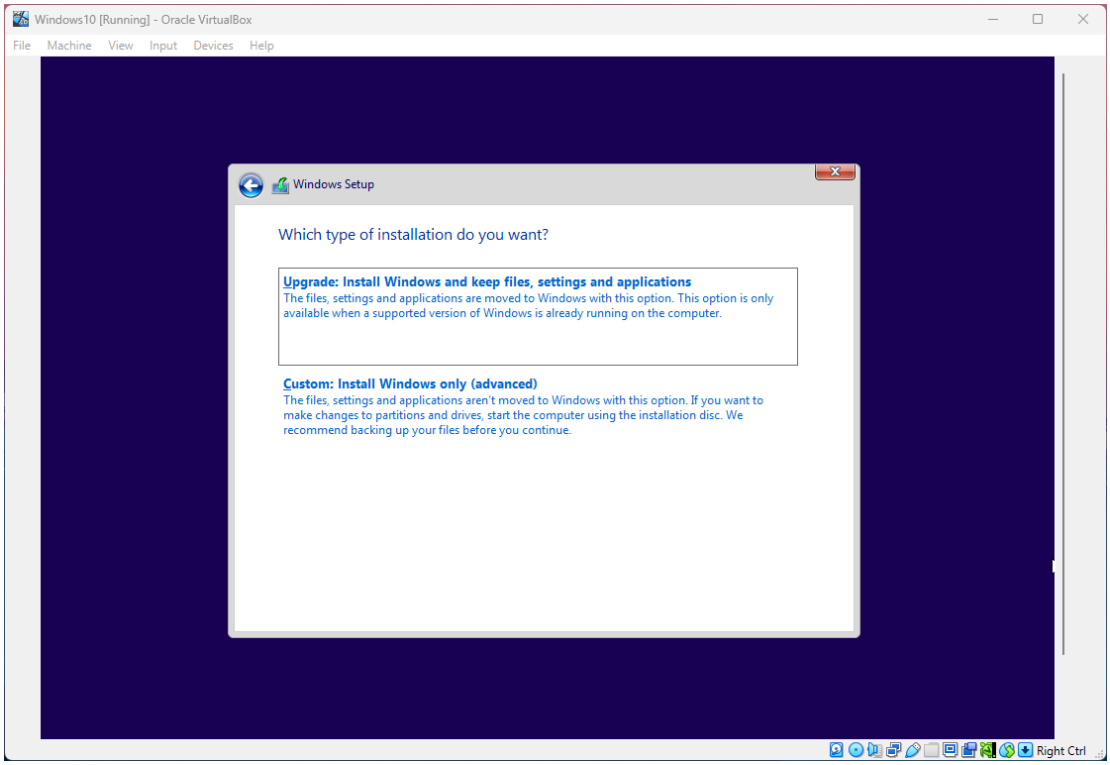
Once the windows 10 boots up, you will be prompted with the first set of installation. Just select options that suit you the best here, and then click next.



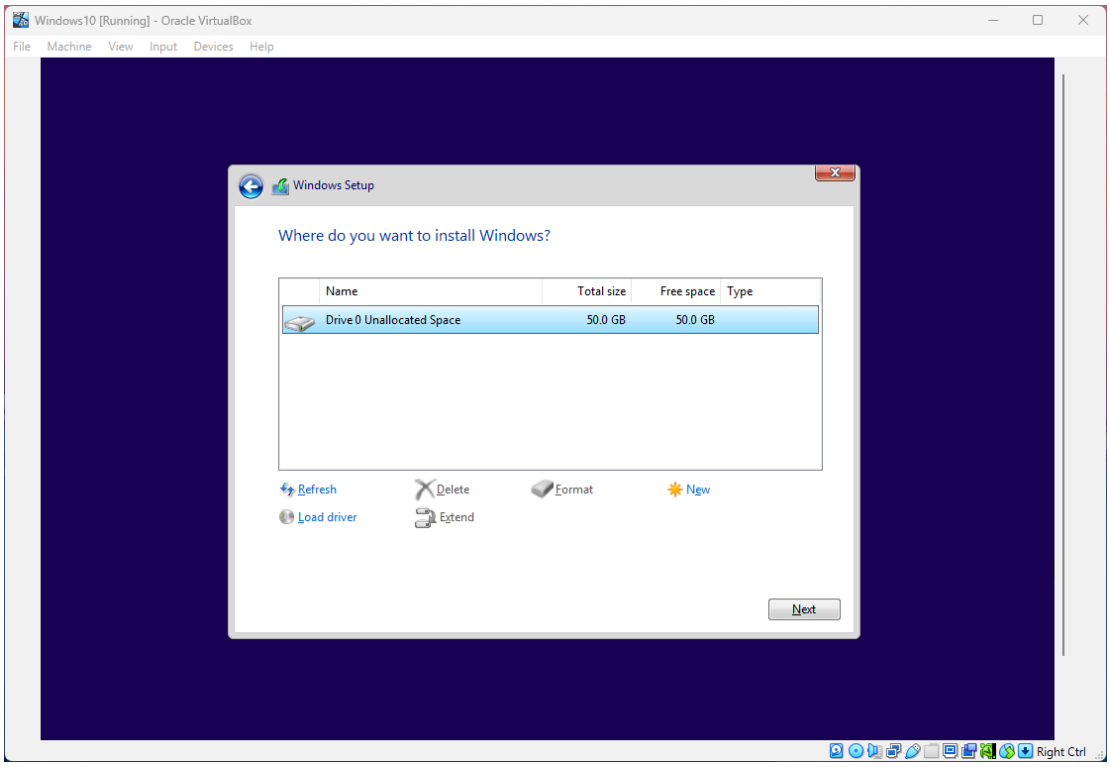
Since I don't have a product key, I will click on "I don't have a product key" option at the bottom, next to button "next".



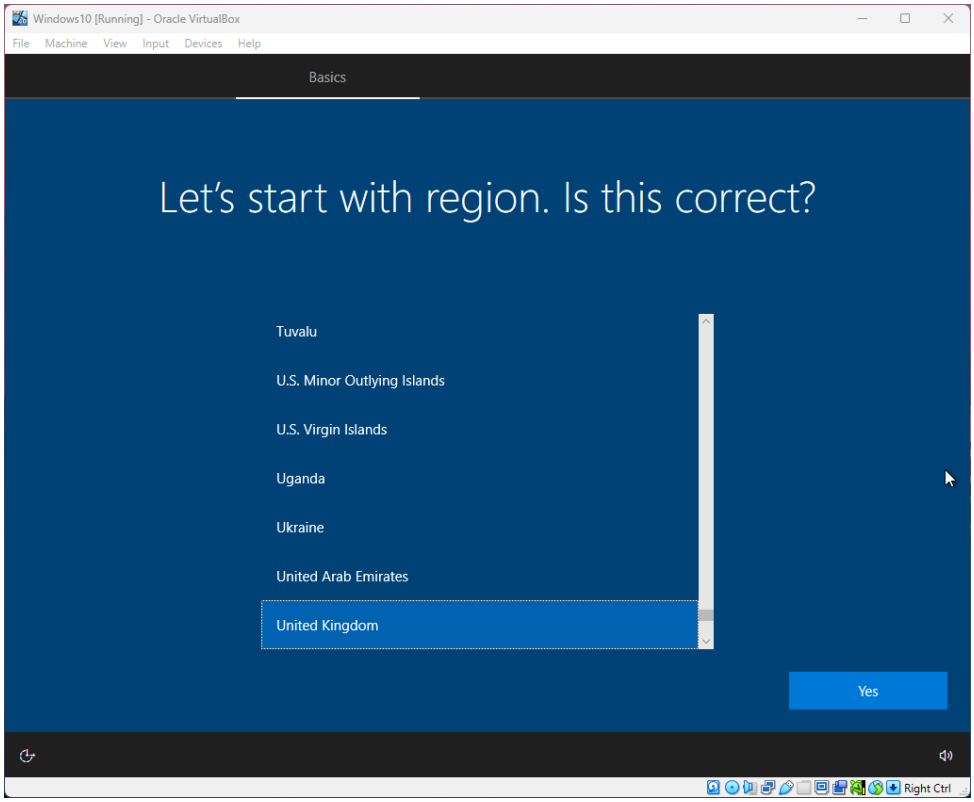
Then it prompts to select a windows 10 version, I will go with windows 10 pro. Then click next.



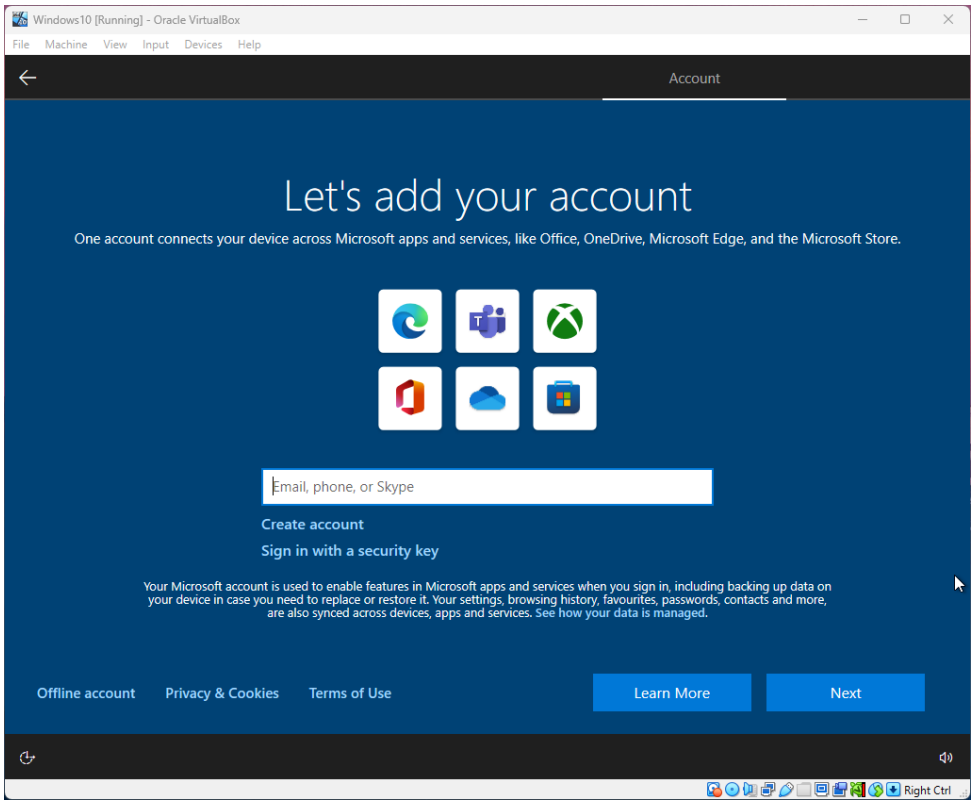
Choose “Upgrade: Install Windows...”



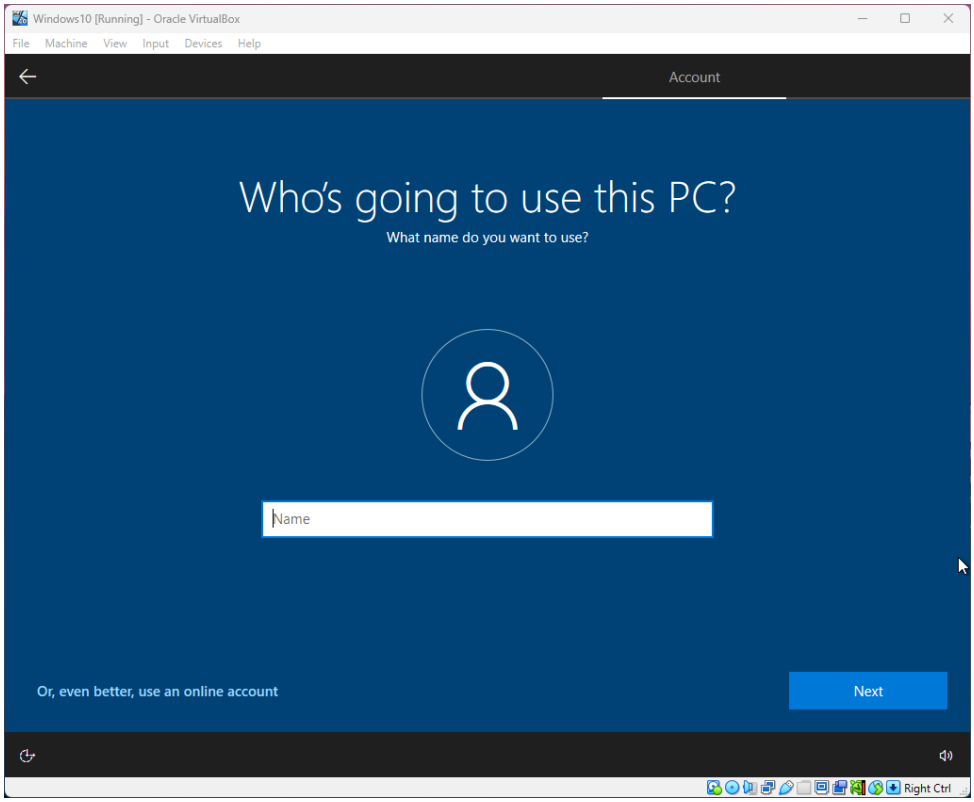
Select the drive we have created at the beginning before we initialised the virtual machine. Once you click next, it will restart and then start the second part of installation.



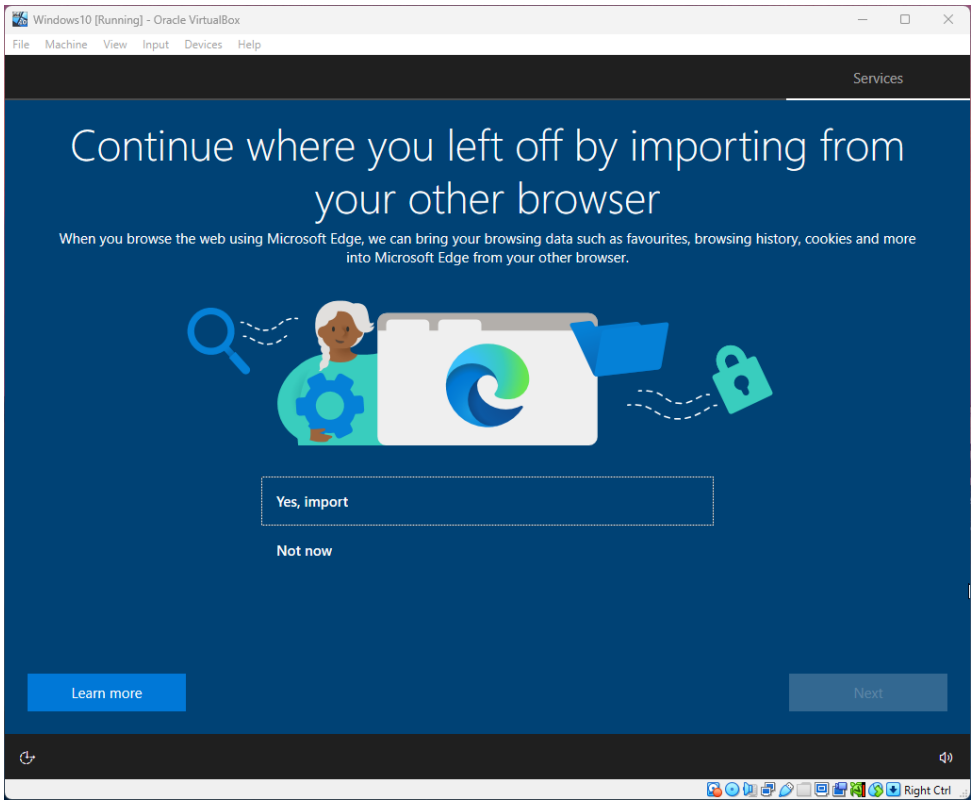
From this point any configurations you make is purely based on what you want.



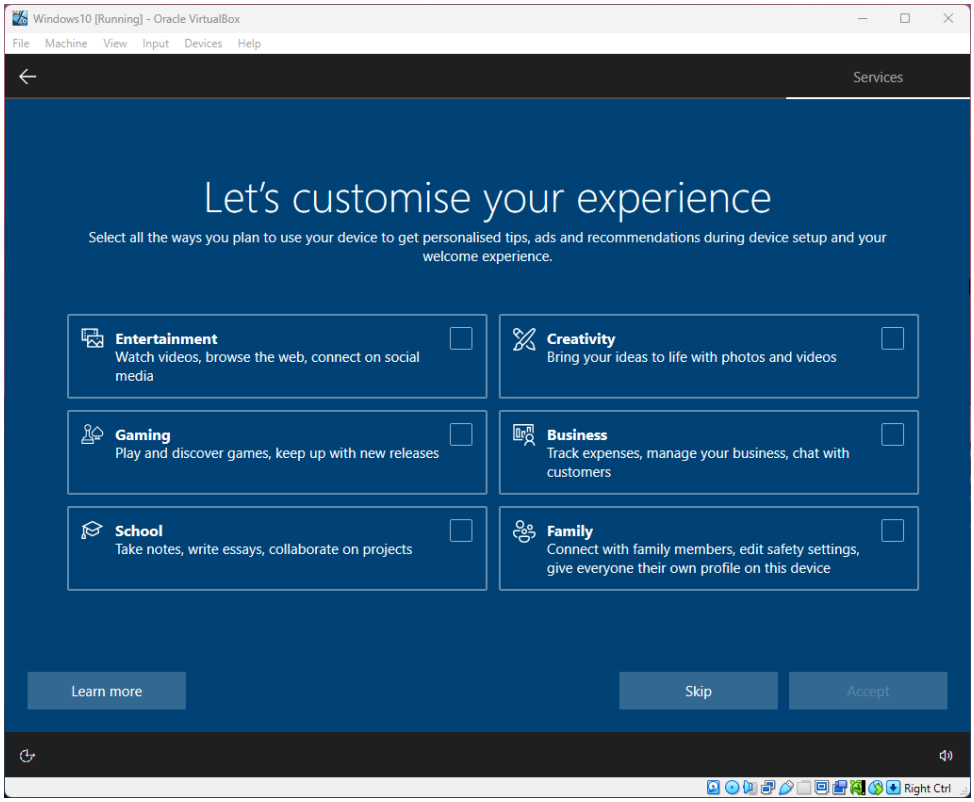
At this step instead of creating an account or using your account, select the offline account option at the bottom left side of the screen. Since we will be using this machine for testing, there is no need for an account to be created.



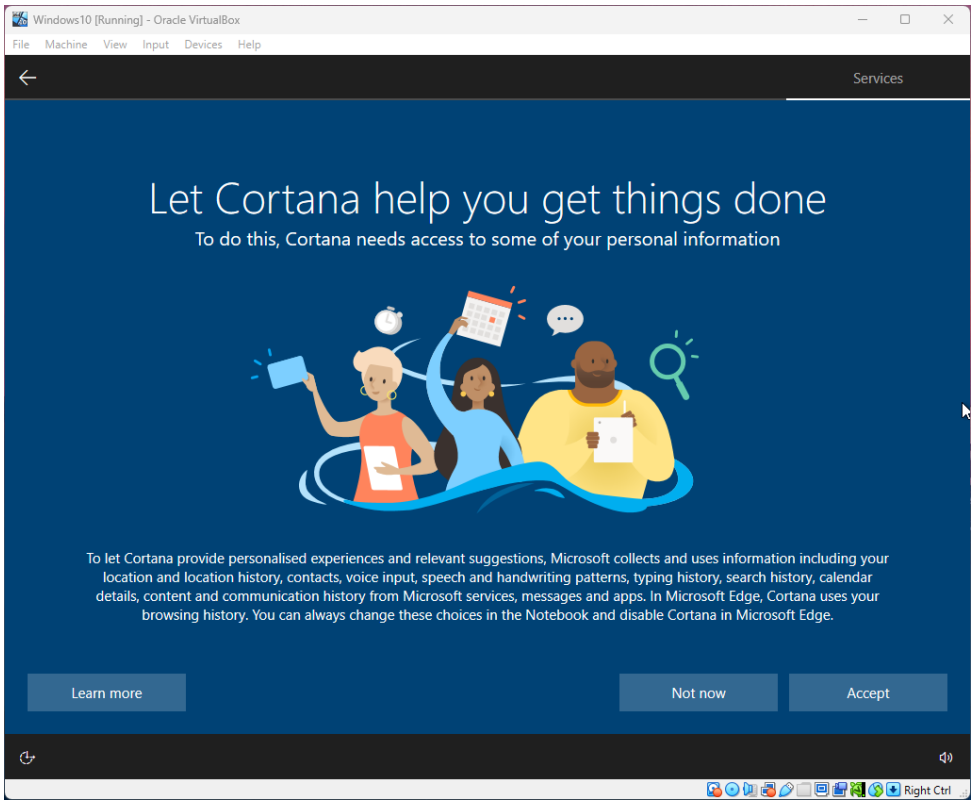
Once all the configuration is done, it will prompt to enter name of an account. Then click next and it will be same for password.



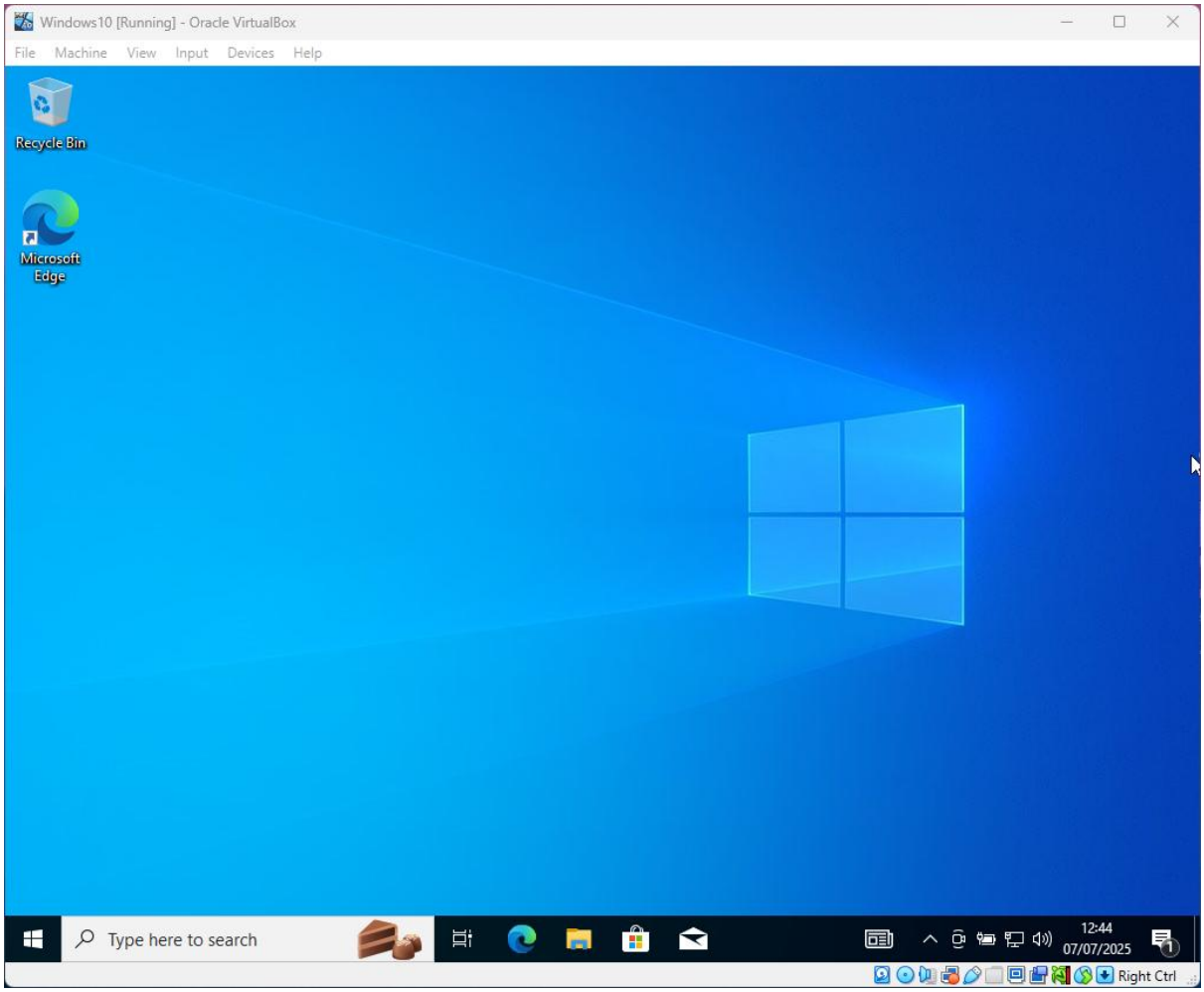
Select “not now” and click next.



Press skip as its optional.

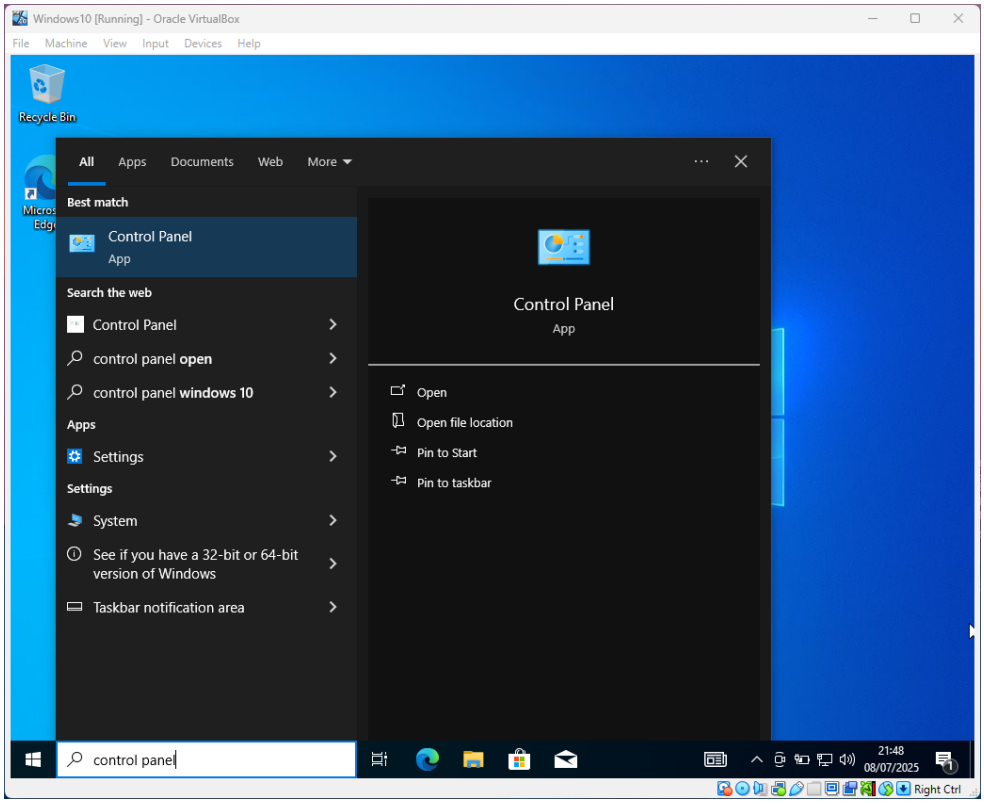


Press not now, unless you want cortana to be enabled.

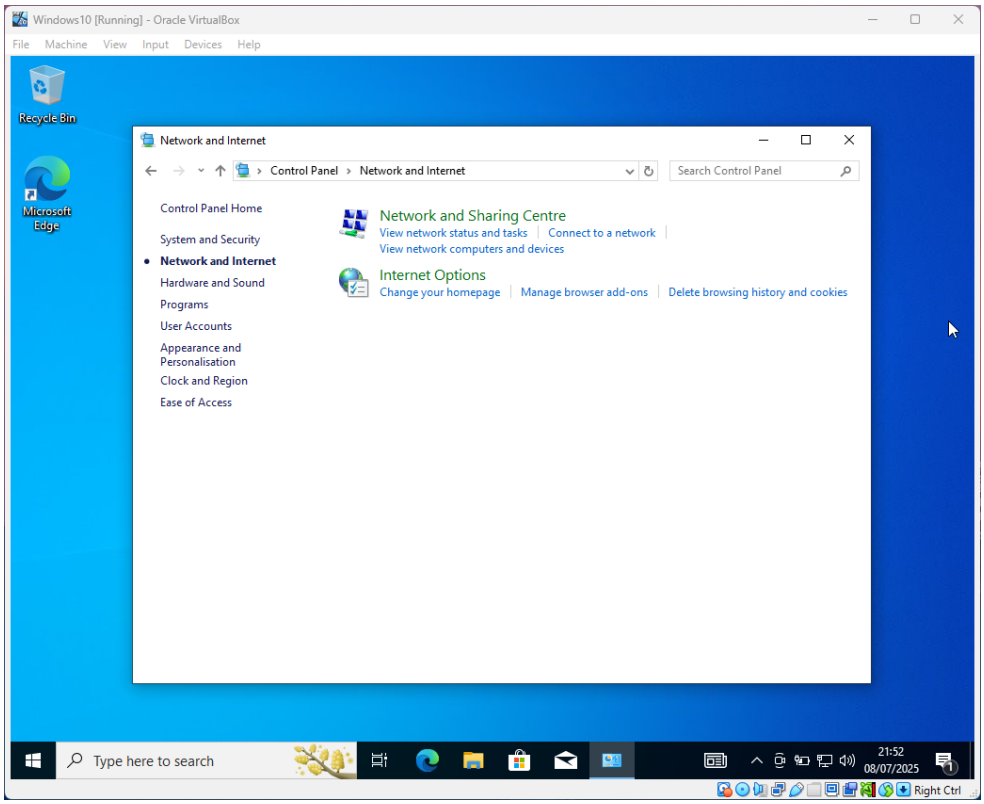


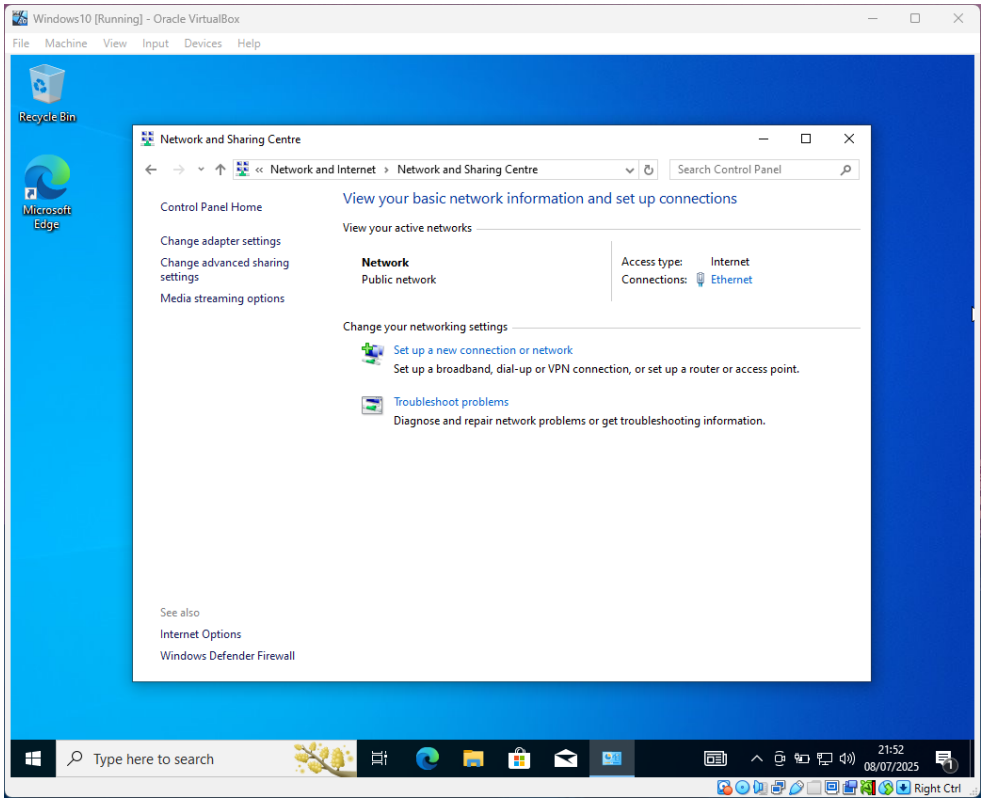
If everything is done correctly, you will be able to log into the windows 10 and see the home screen.

Post Installation Configuration

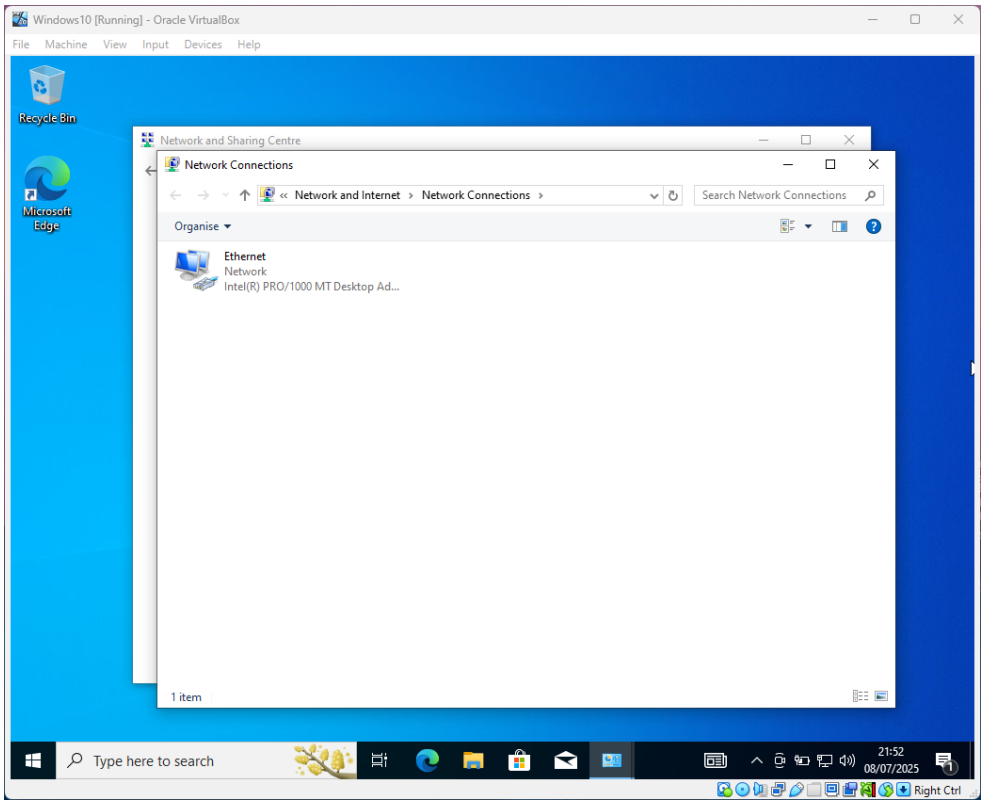


To change the IP address, several steps need to be made. First navigate to Control Panel. Then click on Network and Sharing Centre.

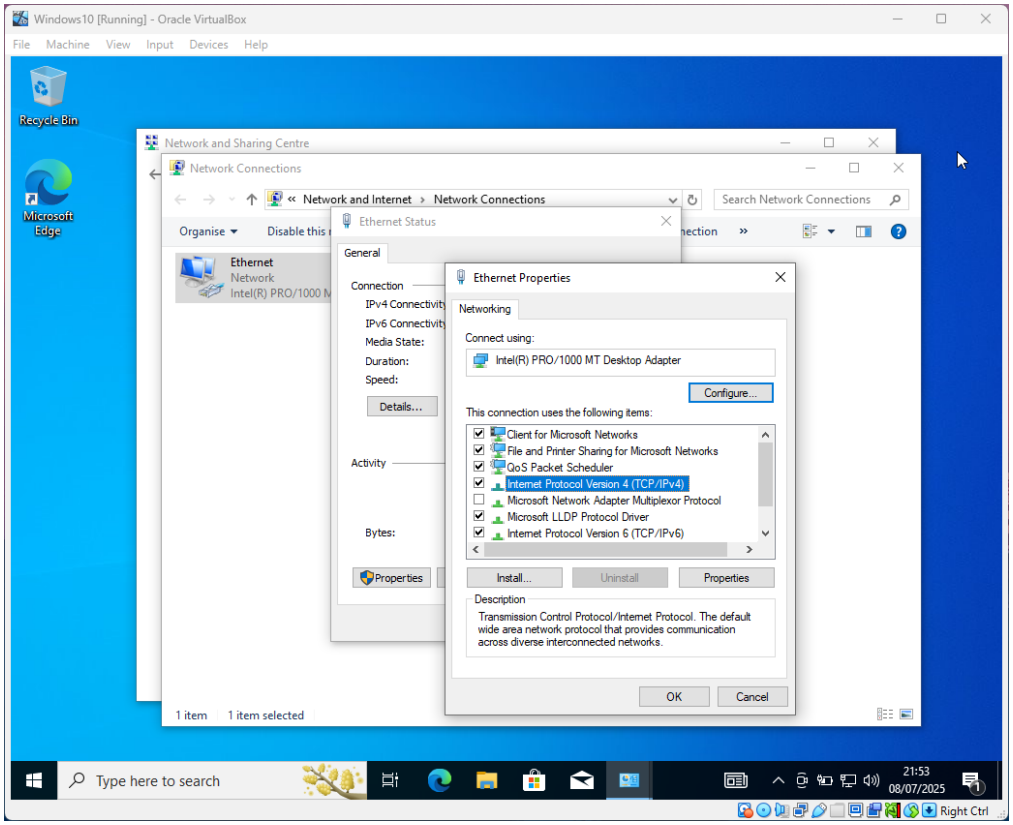




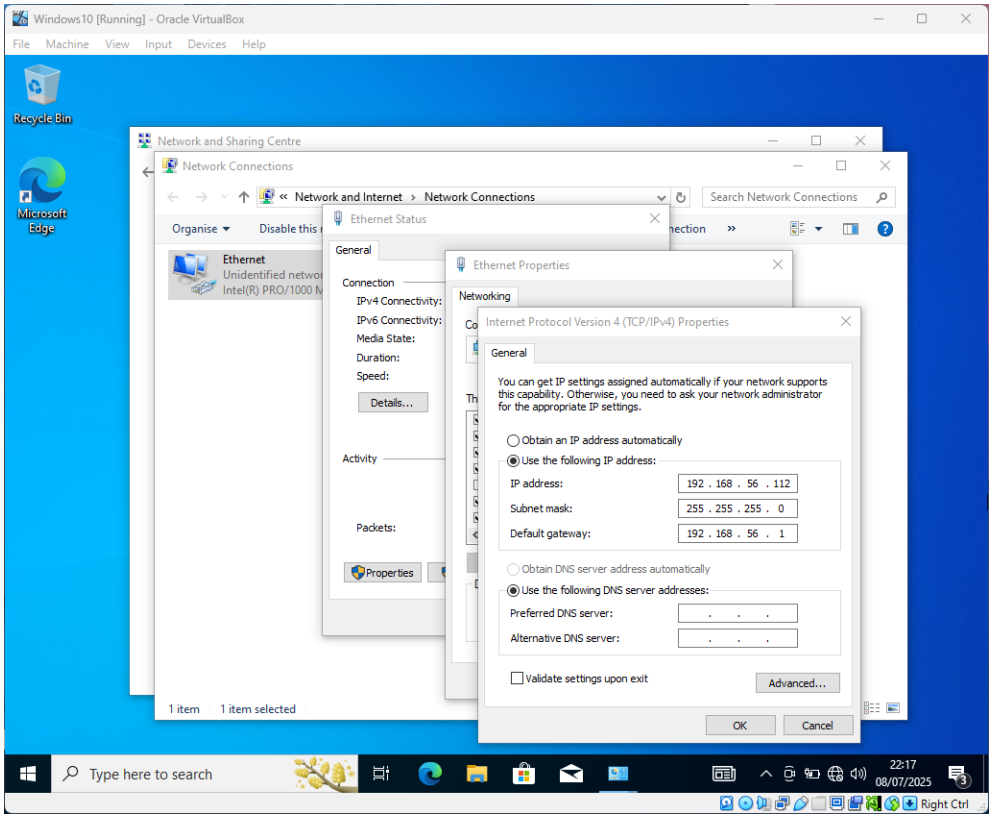
In the network and sharing centre, click on “change adapter settings” on the left side panel.

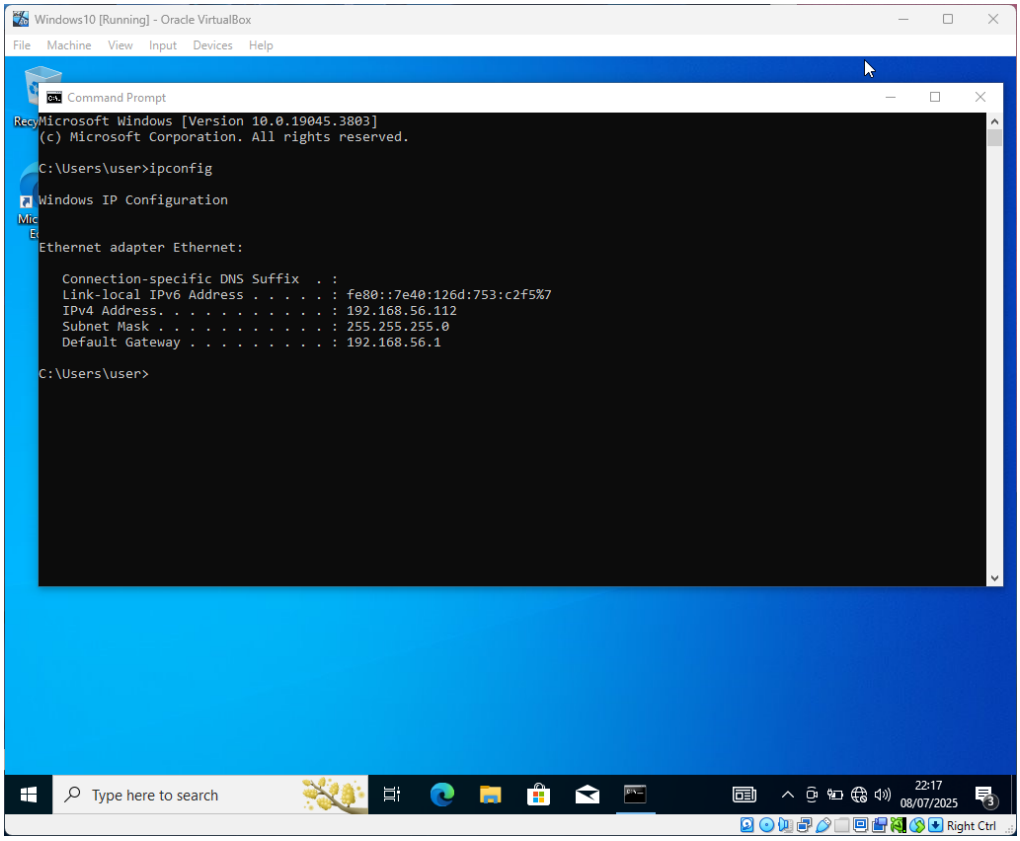


Then click on the ethernet option. This will take you to the configuration window.

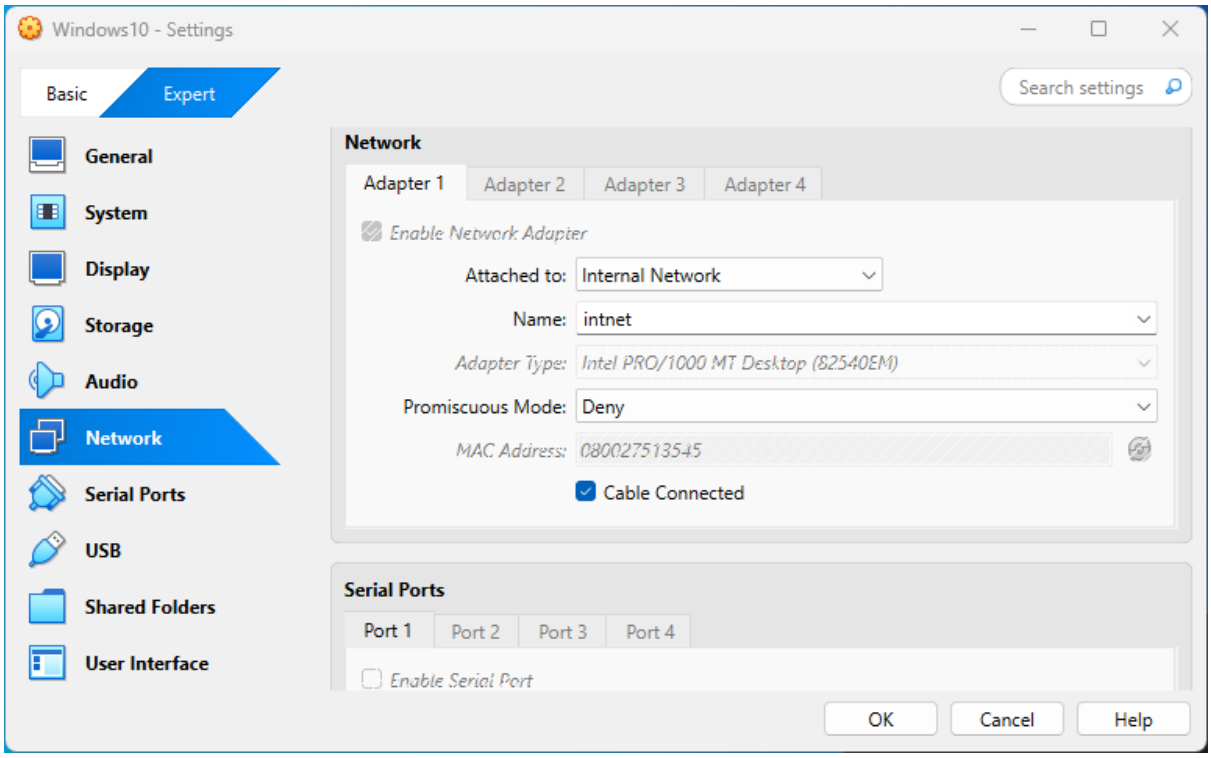


Then go properties and it should open a property window as shown above. Find “Internet Protocol Version 4 (TCP/IPv4)”, select it and click properties. This should lead to a new window, choose use the following IP address and enter your desired IP address.



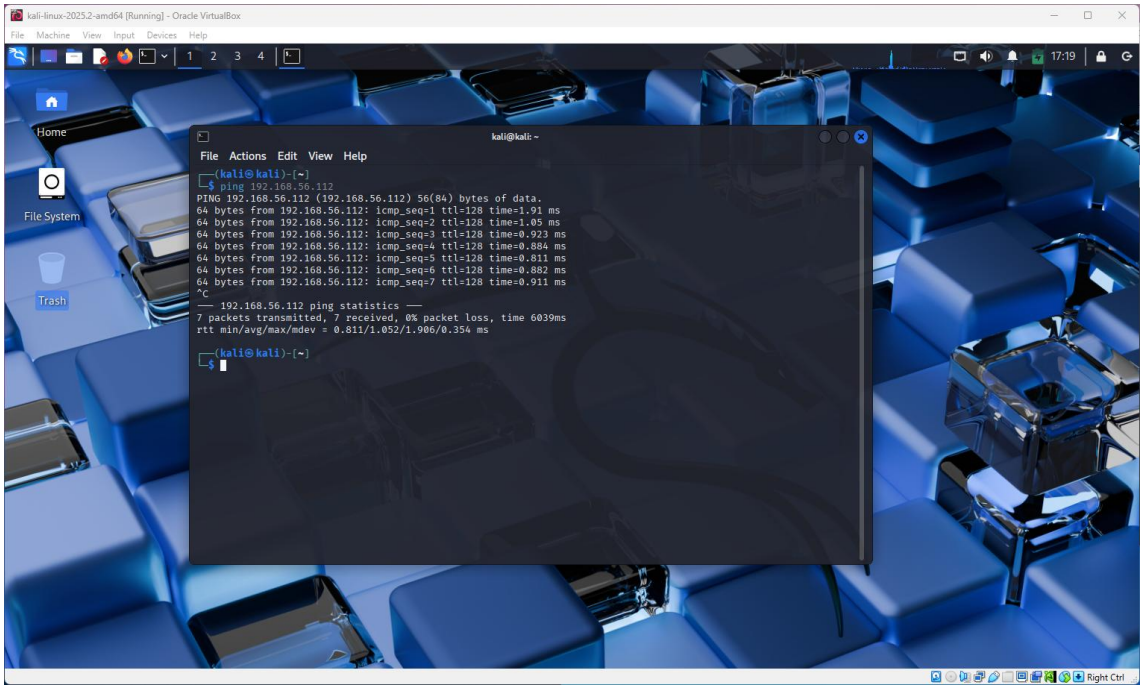


Once IP address is set, open a command prompt. Type “ipconfig” which will show if IP address was changed. In this case it did.

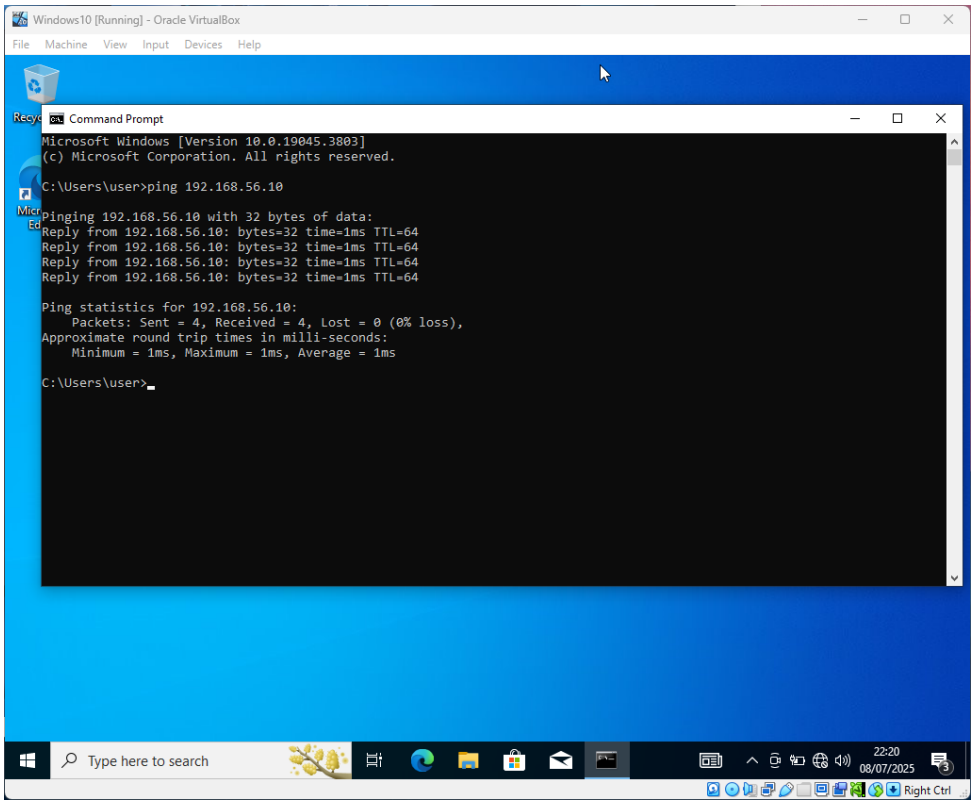


Windows 10 VM settings, go to network settings and change the NAT to Internal Network. Then press OK.

Connection Check



To confirm that the network configuration in Windows 10 was successful, I will run the Kali Linux VM and try to ping the Windows 10 machine. In this example, it connected successfully. If for some reason it shows Host Unreachable then check IP addresses or try to disable Windows Firewall (or allow this IP address to go through).



To test that connection works both ways, I did a ping test on windows to kali VM.

Future Plans / Next Steps

Future plans for this VM are to perform some vulnerability tests, then perform OS hardening. In addition, this machine will be used to generate the traffic to the ubuntu server (which I will setup next) and use different techniques and tools to capture the traffic.

References

Installation Media: https://www.microsoft.com/en-gb/software-download/windows10

Guide on how to install: https://thetechylife.com/how-do-i-create-a-windows-10-iso/

How to setup static IP: https://www.howtogeek.com/19249/how-to-assign-a-static-ip-address-in-windows/